Peizhi Chen

chenpeizhi66@gmail.com | github.com/chenpeizhi66 | linkedin.com/in/peizhi-chen/

Education

Carnegie Mellon University *Pittsburgh, US*

MSc in Computational Design and Manufacturing *Aug 2022 - Feb 2024*

- GPA: 3.7/4.0

Dalian University of Technology *Dalian, China*

Bachelor in Mechanical Engineering *Sept 2018 - June 2022*

- GPA: 3.8/4.0

Research Experience

UAV Autonomous Navigation (Sim-to-Real) *Pittsburgh, PA*

Carnegie Mellon University

- Developed an RRT-based motion planning pipeline for UAV navigation, incorporating dynamic obstacle avoidance and goal-reaching capabilities, and validated the planner on real-world hardware.
- Built a multi-modal perception stack: voxel maps and ray casting for occupancy, plus U-depth, DBSCAN, YOLO, and a Kalman filter to detect and track moving obstacles.
- Set up parallel training and deployment checks in Isaac Sim.

LiDAR Perception for Warehouse Window Localization *Pittsburgh, PA*

Carnegie Mellon University

- Localized window frames in a warehouse model from LiDAR point clouds; object detection reached about 85% accuracy.
- Removed ground points with RANSAC, clustered the remaining cloud with DBSCAN, and classified objects with PointNet++.

Autonomous Navigation for Unitree G1 (Isaac Gym) *Pittsburgh, PA*

Carnegie Mellon University

- Built a simulation navigation stack on Unitree G1, combining LiDAR SLAM and hierarchical planning for goal-directed motion.
- Set up the G1 environment in Isaac Gym with locomotion control and multi-modal sensor streams.
- Ran FAST-LIO2 for real-time LiDAR odometry and mapping, then used TARE Planner (global) and FAR Planner (local) for path planning with dynamic obstacle avoidance.

Industry Experience

Software Engineer, Chip Middleware Team (Nxperf) *Shenzhen, CN*

NIO *Mar 2024 – Present*

- Built **Nxperf**, an NPU profiler in the style of **NVIDIA Nsight Systems**, tracing from operator execution down to hardware scheduling for bottleneck analysis.
- Designed the profiler lifecycle and CLI (launch/profile), and an **LTTng** trace path from collection to parsing to visualization, joining NPU kernel traces with OS scheduling.
- Optimized CVA operators (**concat**, **im2col**, **copy_if**, **quantize**) in **CUDA** with vectorized load/store; up to **40%** faster inference.

Software Engineer, Chip Middleware Team (RDK / ROS 2) *Shenzhen, CN*

NIO *Mar 2024 – Present*

- Built the camera/perception stack in NIO's **Robot Development Kit (RDK)** as **ROS 2** nodes, running on **Ubuntu 24.04** and the in-house chip. Client/server design, stable under multi-process and high concurrency; the same service also ships on the NIO driving stack (**500K+** vehicles).
- Developed a state machine and a modular image pipeline for cross-process and chip-to-chip traffic (**IPC / C2C**); control plane is **gRPC + Protobuf** for multi-client access.
- Deployed a **vision-language-action (VLA)** model as ROS 2 nodes on the same SoC, and brought up **ROS-LLM** for language-conditioned robot control on the RDK.

Technical Skills

Programming Language C/C++ 17, Python, Matlab

Tools and frameworks PyTorch, Cuda, Numpy, Scipy, Tensorflow, Git, Latex, OpenCV, PCL

Environment Linux, ROS2, GNU Bash, Conda, Vim, Docker